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Cascading of circular business models in the textile and clothing industry: Insights for operationalization through a systematic review

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ABSTRACT

The circular economy seeks to preserve resource value by promoting circular business models (CBMs) such as repair, reuse, remanufacturing, and recycling. However, the practical operationalization of cascading – where resources are sequentially utilized across multiple CBMs to maximize their residual value – remains insufficiently explored, particularly in the textile and clothing (T&C) industry. This sector, marked by high resource consumption, short product lifespans, and significant environmental impacts, presents unique opportunities to slow resource loops and enhance value creation. This study systematically reviews 45 academic articles to examine how cascading is operationalized within the T&C sector. Seven archetypes of cascading are identified, encompassing interactions within and between inner loops (e.g., rental, repair, reuse), outer loops (e.g., remaking, recycling), and open-loop applications across value chains. Using cascade chain theory, these practices are mapped onto four guiding principles: appropriate fit, augmentation, consecutive relinking, and balancing resource metabolism. The analysis reveals key enabling factors for effective cascading, including advanced sorting technologies, digital and physical tracking systems, circular design, life extension strategies, supply chain coordination, supportive policies, eco-material substitution, and innovative recycling techniques. Despite these enablers, scaling cascading operations is hindered by infrastructural limitations, market and safety challenges, insufficient local processing capacity, and regulatory barriers. The findings highlight critical factors shaping cascading practices and emphasize the need for coordinated action among T&C stakeholders, policymakers, and intermediaries to establish robust cascade chains and advance the transition to circularity.

1. Introduction

In recent years, rising consumption and the premature disposal of products in landfills have posed significant threats to environmental sustainability (Alves et al., 2024). In response, the circular economy (CE) has gained momentum, increasingly shaping public policies and corporate strategies (Campbell-Johnston et al., 2020). Central to the CE framework are R-imperative circular business models (CBMs) – such as reuse, refurbishment, and recycling – which aim to retain product integrity and preserve resource value (Reike et al., 2018). These models contribute to slowing and closing resource loops (Bocken et al., 2025), while also fostering broader systemic change by influencing regulatory frameworks and promoting social transformation (Han et al., 2023; Klofsten et al., 2024).

Building on the role of CBMs in enabling systemic change, the concept of cascading has been introduced as a structuring mechanism within the broader CE paradigm, aligned with its value retention hierarchy (Campbell-Johnston et al., 2020; Ellen MacArthur Foundation, 2017; Lüdeke-Freund et al., 2018). Originally conceptualized by Sirkin and ten Houten (1994), cascading embodies a systemic logic that organizes CBMs – such as repair, reuse, refurbishment, remanufacturing, and recycling – according to the progressive decline in resource quality or functionality. This sequential structure, likened to *"a river flowing over a sequence of plateaus"* (ibid., p. 215), ensures that resources are utilized to their fullest potential at each stage before transitioning to lower-value applications. Importantly, cascading not only guides the technical flow of materials but also facilitates the institutional embedding of resource (re)circulation, coordinating actions across diverse actors, policy frameworks, and socio-technical practices (Campbell-Johnston et al., 2020). As inferred by Lata et al. (2023), this institutional dimension is

Abbreviations: CE, Circular Economy; CBM, Circular Business Model; CCT, Cascade Chain Theory; T&C, Textile and Clothing; SLR, Systematic Literature Review.

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essential for reconciling competing economic, social, and ecological priorities within fragmented CE transition landscapes. From this perspective, cascading emerges as a promising mechanism for institutionalizing 'systemic circularity', linking material flow governance with institutional coordination to advance value retention and resource conservation.

The premise of this paper is that while cascading has gained traction in CE discourse, its operationalization remains fragmented and underexplored, particularly in technical cycles involving durable goods such as electronics, textiles, and machinery. Existing studies predominantly focus on biological cycles, such as timber and food waste (Besserer et al., 2021; Taskhiri et al., 2016; Zabaniotou and Kamaterou, 2019), where material flows are relatively linear and well understood. In contrast, technical cycles – characterized by complex product compositions, high-value materials, and embedded energy – offer significant but underexamined opportunities for value retention through CBMs such as repair, reuse, refurbishment, and remanufacturing (Hopkinson et al., 2018). Cascading could be especially impactful in these contexts by enhancing resource conservation and reducing emissions. However, this potential is constrained by systemic challenges, including reverse logistics (Cichosz et al., 2025), lack of design for circularity (Wang et al., 2022), and fragmented stakeholder coordination (Dagilienė and Varaniūtė, 2023). As a result, cascading is often conflated with closed- and open-loop recycling (Campbell-Johnston et al., 2020), overlooking CBMs with higher priority in the CE value hierarchy and their role in systemic circularity. Crucially, no integrative framework exists to align cascading with the CE value hierarchy in order to guide decision-making, particularly in relation to value flows, value capture, and coordination across actors and organizations (Campbell-Johnston et al., 2020; Lüdeke-Freund et al., 2018). This shortfall is especially evident in circular supply chain research, where empirical insights into collaborative mechanisms and cascading loops remain limited (Aarikka-Stenroos et al., 2022). This paper addresses these gaps through a systematic literature review, aiming to synthesize existing knowledge, identify blind spots, and propose pathways for institutionalizing cascading in technical cycles through design, policy, and business innovation.

The textile and clothing (T&C) industry offers a particularly relevant context for examining the operationalization of cascading within CE technical cycles. Characterized by complex material flows, high resource consumption, and low recovery rates (Alves et al., 2024; Charnley et al., 2024), the industry exemplifies both the challenges and opportunities associated with cascading. A striking example is clothing disposal: on average, each European discards around 11 kg of clothing annually, with nearly 87 % ending up in landfill or incineration (European Parliament, 2020). This scale of waste contributes to the sector's reputation as one of the most polluting and resource-intensive globally (Alves et al., 2024) and has led to its prioritization within the EU's CE agenda (European Commission, 2022). Although CBMs such as rental, repair, resale, remaking, and recycling are increasingly practiced in this sector (Dehghannejad et al., 2025; Salmi and Kaipia, 2022), their integration into a cascading framework remains underdeveloped. Addressing this operational challenge requires a holistic perspective – one that considers enabling conditions such as design strategies for repair and upgradability, data traceability systems, collaborative business models, and supportive financial mechanisms (Bocken et al., 2016; Charnley et al., 2024), alongside organizational, regulatory, and technological barriers (Brändström et al., 2024) that may isolate these models. Collectively, these conditions make the T&C industry a compelling case for investigating how cascading can enhance the valorization of products and materials, improve resource flows, and support systemic circularity.

Accordingly, this paper, through an SLR, explores cascading operations in the T&C industry by addressing the following research questions:

RQ1. How is cascading operationalized within the T&C industry?

RQ2. How do underlying facilitators and barriers affect cascading operations in this sector?

To address these RQs, we infuse the cascade chain theory (CCT) (Sirkin and ten Houten, 1994), as it builds on the concept of cascading, in which materials and resources are sequentially reused to maximize their utility by leveraging their remaining quality at each stage, following four key principles, as detailed in Section 2. This approach enhances the understanding of valorizing resources across circular loops, including the factors that facilitate or hinder cascading operations, and helps translate cascading of CBMs into practical applications in the resource-intensive T&C industry.

The structure of this paper is as follows: Section 2 introduces the theoretical framework that underpins the study. Section 3 outlines the research method. Section 4 analyzes the literature and cascading archetypes, identifying key facilitators and barriers to cascading operations. Section 5 discusses the main findings, culminating in four dimensions that shape cascading in the T&C industry. Finally, Section 6 presents the conclusions.

2. Theoretical framework

2.1. CE, CBM, and cascading in the T&C industry context

CE offers a transformative paradigm for sustainable development, particularly relevant to the T&C industry. Defined as *"a system where materials never become waste and nature is regenerated"* (Ellen MacArthur Foundation, 2025), CE challenges the dominant linear "take, make, dispose" model by advancing regenerative, closed-loop systems (Lopes de Sousa Jabbour et al., 2019). Its core principles – eliminating waste and pollution, keeping products and materials in use, and regenerating natural systems – are operationalized through two interconnected cycles: biological and technical (Ellen MacArthur Foundation, 2019). In the biological cycle, circularity is achieved through the design of materials that can safely return to nature, exemplified by biodegradable fibers and bio-based recycling of cotton waste (Grover et al., 2025), though still limited in scale. By contrast, the technical cycle, which dominates the T&C sector, pursues circularity through business models that retain the embedded value of manufactured goods, conserving energy, materials, and capital (Hopkinson et al., 2018).

Within this technical cycle, CBMs play a central role in achieving CE goals. In the T&C context, CBMs encompass strategies such as repair, rental, resale through take-back schemes, swapping, upcycling/redesign, and mechanical or chemical recycling (Ellen MacArthur Foundation, 2021). These CBMs contribute to circularity by either slowing loops – extending product lifespans – or closing loops – recovering material value (Bocken et al., 2016; Brenner and Drdla, 2023). Organized across inner and outer loops, they form a hierarchy of value retention (Reike et al., 2018). Inner-loop practices, such as reuse, repair, and rental, extend the life of garments while preserving their integrity and function (Bakker et al., 2019), as seen in platforms like Rent the Runway (Pal and Gander, 2018). Outer-loop practices, including remanufacturing and fiber-to-fiber recycling technologies such as Circulose (Circulose, 2025), involve disassembly and revalorization, generating new production cycles when garments can no longer be reused in their original state (Bakker et al., 2019; Ellen MacArthur Foundation, 2022). Together, these loops illustrate how CE maximizes longevity and preserves material value across lifecycles.

While CBMs define the strategies through which circularity is pursued, a critical mechanism that connects and extends these loops is cascading (Campbell-Johnston et al., 2020). Cascading refers to the sequential use of resources across different CBMs, aiming to extract maximum value from their material and energy integrity (Ellen MacArthur Foundation 2024a, 2024b; Sirkin and ten Houten, 1994). In the

T&C context, for example, a garment may first be worn as second-hand clothing, then transformed into new textile products, and eventually repurposed into industrial applications such as insulation or composite paper (Sirkin and ten Houten, 1994; Wang et al., 2023). Hybrid initiatives such as ACS Clothing, which integrates rental, resale, and repair strategies to keep garments circulating in inner loops, or Patagonia's Worn Wear, which combines repair and resale with outer-loop CBMs like recrafting and recycling, demonstrate how cascading enables garments to progress through successive pathways (Circular X, 2020; Ellen MacArthur Foundation, 2024a). These multi-stage transitions maximize resource efficiency and minimize waste.

However, not all circular activities qualify as cascading. For an activity to be considered cascading, products or materials must enter the reverse supply chain after their initial use phase, involving a shift in usership/ownership, or application at the end-of-use or end-of-life (Campbell-Johnston et al., 2020; Kalverkamp et al., 2017; Sirkin and ten Houten, 1994). For instance, repair-based CBMs only count as cascading when repaired garments are transferred to new users; if the item remains with the original owner, its lifespan is simply extended in a linear fashion, without enabling cascading (Sirkin and ten Houten, 1994). The same applies to rental-based CBMs: if the model is not implemented as part of a reverse supply chain, the change in usership alone does not constitute a cascaded operation. This distinction differentiates cascading-related CBMs from other circular practices.

2.2. CCT

CCT can be analogically understood through the descent of a river across plateaus, where water transforms into whirlpools, foam, or vapor before settling in a lake, symbolizing the transformation of material and energy (Sirkin and ten Houten, 1994). This illustrates how resources undergo multiple stages of use, often beyond their original function, reflecting the dynamic nature of circular flows. As depicted in Fig. 1, resources may follow a primary cascade chain, transitioning from "Use 1" to "Use n" as their quality gradually declines over time, or divert into secondary cascade chains that enable further reuse and value retention (e.g., through different value chains). CBMs play a pivotal role in facilitating these transitions by extending product lifespans and capturing residual value (Campbell-Johnston et al., 2020).

The cascade chain – aligned with the core principles of the CE – supports both top-down and bottom-up flows (Olsson et al., 2018; Sirkin and ten Houten, 1994), operating within short- and long-loop systems and enabled through interactions among stakeholders via exchange channels such as B2B, B2C, or C2C (Reike et al., 2018). However, as resources progress through the cascade, their material order declines due to increasing entropy, highlighting the thermodynamic limitations of circularity (Ellen MacArthur Foundation, 2024a, 2024b). This systemic perspective offers a robust framework for understanding how circular strategies can be optimized to preserve value while acknowledging inevitable material degradation.

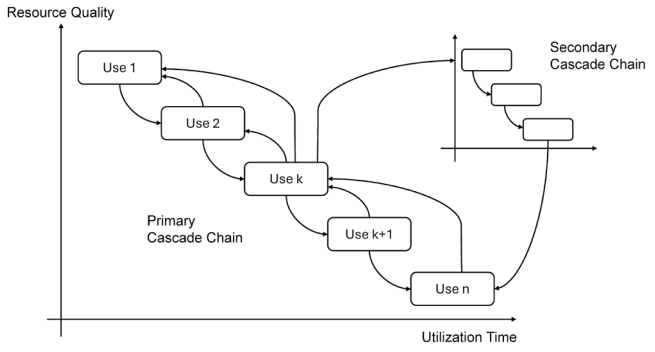


Fig. 1. The cascade chain – adapted from Sirkin and ten Houten (1994, p. 217).

At its core, CCT is based on four dimensions: resource quality, utilization time, salvageability, and consumption rate, which respectively reflect four key principles (Sirkin and ten Houten, 1994):

- Appropriate Fit: Matching resource quality to task difficulty – that is, using high-quality resources for demanding tasks and lower-quality ones for simpler tasks.
- Augmentation: Enhancing resource quality or durability through repair, reuse, or protective treatments to extend lifespan.
- Consecutive Relinking: Salvaging and reintegrating resources into new processes after each use, thereby turning the previous two principles into a cohesive cascade chain.
- Balancing Resource Metabolism: Regulating resource consumption and generation to avoid depletion and support long-term sustainability.

As Sirkin and ten Houten (1994) highlight, while the first three principles address operational efficiency, the fourth introduces a normative layer, emphasizing the importance of responsible resource management in both policy and practice.

3. Method

To explore how cascading is operationalized in the T&C industry, an SLR is strategically deployed for several reasons. First, due to the complexity and evolving nature of CE practices in T&C, an SLR offers a rigorous, transparent, and replicable approach to answering targeted research questions (Denyer and Tranfield, 2009; Snyder, 2019), which is essential for understanding cascading and identifying its facilitators and barriers. Second, cascading in this context is an emerging, transdisciplinary concept spanning environmental, economic, and design fields. An SLR enables the synthesis of dispersed knowledge, revealing credible patterns and missing links (Denyer and Tranfield, 2009). Additionally, it enhances transferability by aggregating knowledge across contexts, allowing readers to assess its relevance to their own settings. Finally, SLR integrates theoretical perspectives with empirical findings, providing a more comprehensive view of cascading operations to guide future research and policy development. It also improves results validity by avoiding duplicate analyses and reducing bias (Denyer and Tranfield, 2009; Wolfswinkel et al., 2017). This review begins by applying predefined selection criteria and conducting a content analysis to examine cascading archetypes. Building on this foundation, and guided by the CCT's four principles, it then identifies the key facilitators and barriers that influence cascading operations.

3.1. Screening and selection of articles

As illustrated in Fig. 2, the SLR design followed the PRISMA guidelines (Page et al., 2021) and utilized two major academic databases: Web of Science and Scopus. To collect data, diverse search criteria and Boolean strings were applied to capture cascading interactions within and between circular loops. The figure provides a visual overview of the screening and selection process, supporting the methodological transparency of the review.

The initial search string, referred to as String 1 – [(cascad*) AND (textile* OR cloth* OR garment* OR apparel*)] – was intended to identify literature explicitly discussing cascading operations in the T&C sector. This search retrieved 366 articles across both databases, but few were relevant, as many focused on unrelated topics such as electrical cascades in smart textiles. Recognizing that cascading operations are closely linked to the CE paradigm and involve various CBM interactions, we refined our approach with String 2 – [("circular economy") AND (textile* OR cloth* OR garment* OR apparel*)] – which yielded 1470 articles from both sources and provided a broader and more relevant dataset aligned with the research aim. In addition, to capture the significance of allied closed-loop processes and cross-sectoral cascading

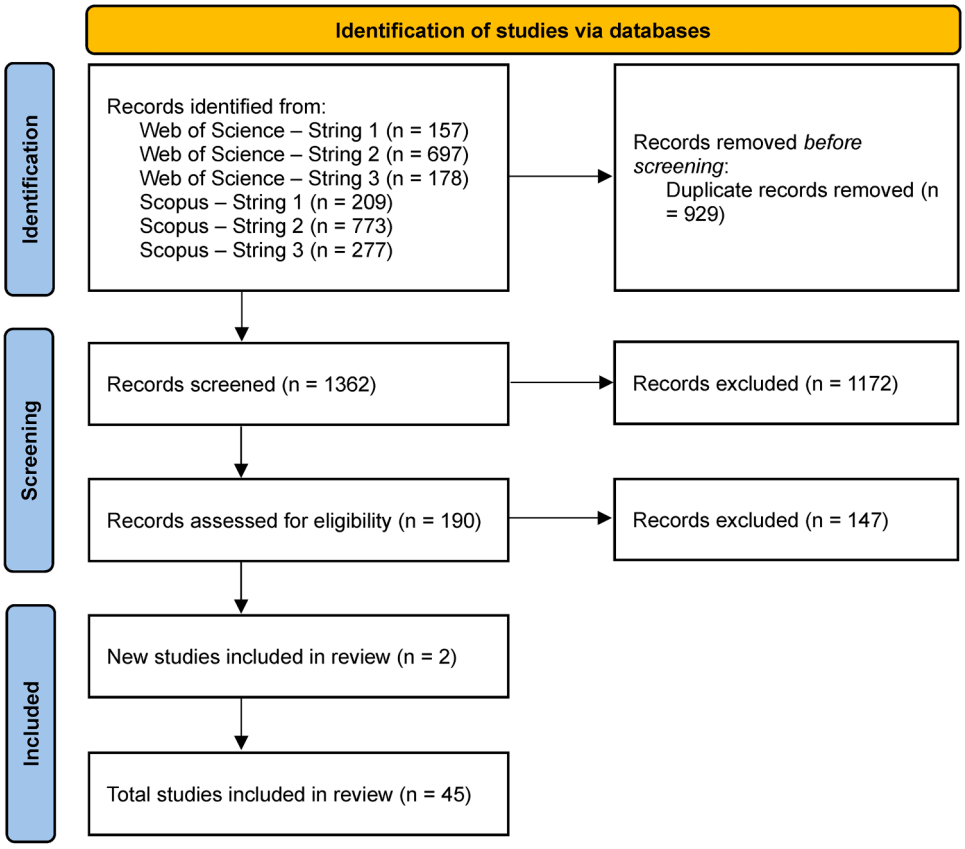


Fig. 2. PRISMA flow diagram of the study selection process.

potential, we conducted an extra search using String 3 – [("closed-loop" OR "open-loop") AND (textile* OR cloth* OR garment* OR apparel*)] – which resulted in 455 articles overall. All searches spanned the full spectrum of academic disciplines and included publications available up to April 2024. Results were filtered by title, abstract, and keywords, focusing on English-language full-length and review articles. This process yielded a total of 2291 articles.

After removing duplicates, 1362 articles remained. The first author screened them for relevance to cascading practices within the technical cycles of the CE. In cases of ambiguity, articles were reviewed collaboratively with co-authors to enhance the validity of the dataset, resulting in 190 eligible articles. The authors then jointly finalized those fitting the cascading definition. As Snyder (2019) states, *"the aim of a systematic review is to identify all empirical evidence that fits the pre-specified inclusion criteria to answer a particular research question"* (p. 334). Therefore, inclusion/exclusion criteria were followed for further screening. Articles were excluded if they: (1) described an isolated circular practice without addressing intra- or inter-sectoral interactions aligned with the cascading framework; (2) focused on rental or repair CBMs limited to primary consumption, without links to secondary use phase or ownership/usership transfer (e.g., basic repair/maintenance that restored the product to its original capacity); (3) presented only hypothetical models, preliminary lab studies, or tech demos, lacking real-world cascading operations or potential for scalable application; or (4) were reviews or generally discussed CE barriers/facilitators without specific relevance to cascading, as they did not provide a detailed description necessary for conducting thematic analysis. This process excluded 147 articles and added 2 via reference screening, resulting in a final set of 45.

3.2. Analytical criteria for content analysis

The analysis alternated between data, themes, and concepts. It initiated inductively to identify cascading operations, followed by an abductive approach to examine underlying facilitators and barriers. As Gioia et al. (2012) emphasize, this strategy minimizes bias by avoiding premature entry into the research literature, while enabling integration of data and theory in the second level of the analysis.

The analytical process, summarized in Fig. 3, began with re-reading the selected articles to extract and synthesize content on cascading operations. The first author used Excel to identify specific operations and develop first-order concepts, which were refined using the "cut and sort" technique (Ryan and Bernard, 2016) to support further coding. This led to the development of second-order themes and aggregated dimensions – referred to as cascading archetypes – by comparing patterns to avoid misinterpretations, particularly around closed/open loops and remaking/remanufacturing activities. Clarification was essential due to frequent confusion between terms such as upcycling and downcycling. The process involved all authors and was inspired by Gioia et al.'s (2012) approach.

The second level of analysis aimed to identify key facilitators and barriers to cascading operations, using content analysis conducted by the first author. Guided by the CCT lens, cascading operations described in each article were examined and mapped against four fundamental cascading principles: appropriate fit, augmentation, consecutive relinking, and balancing resource metabolism (see Table A.1 in the Appendix). For example, processes such as product repair or material treatment using various methods were classified under augmentation, as they aim to enhance quality and extend garment life. Following this mapping, an inductive analysis was performed to identify facilitators and barriers, which appeared in diverse forms. For instance, sorting technologies supported appropriate fit by matching used clothing with suitable end uses, such as reuse or recycling. Likewise, specific services, policies, or technologies enabled the relinking of products and materials into consecutive loops. Barriers were also identified using the same

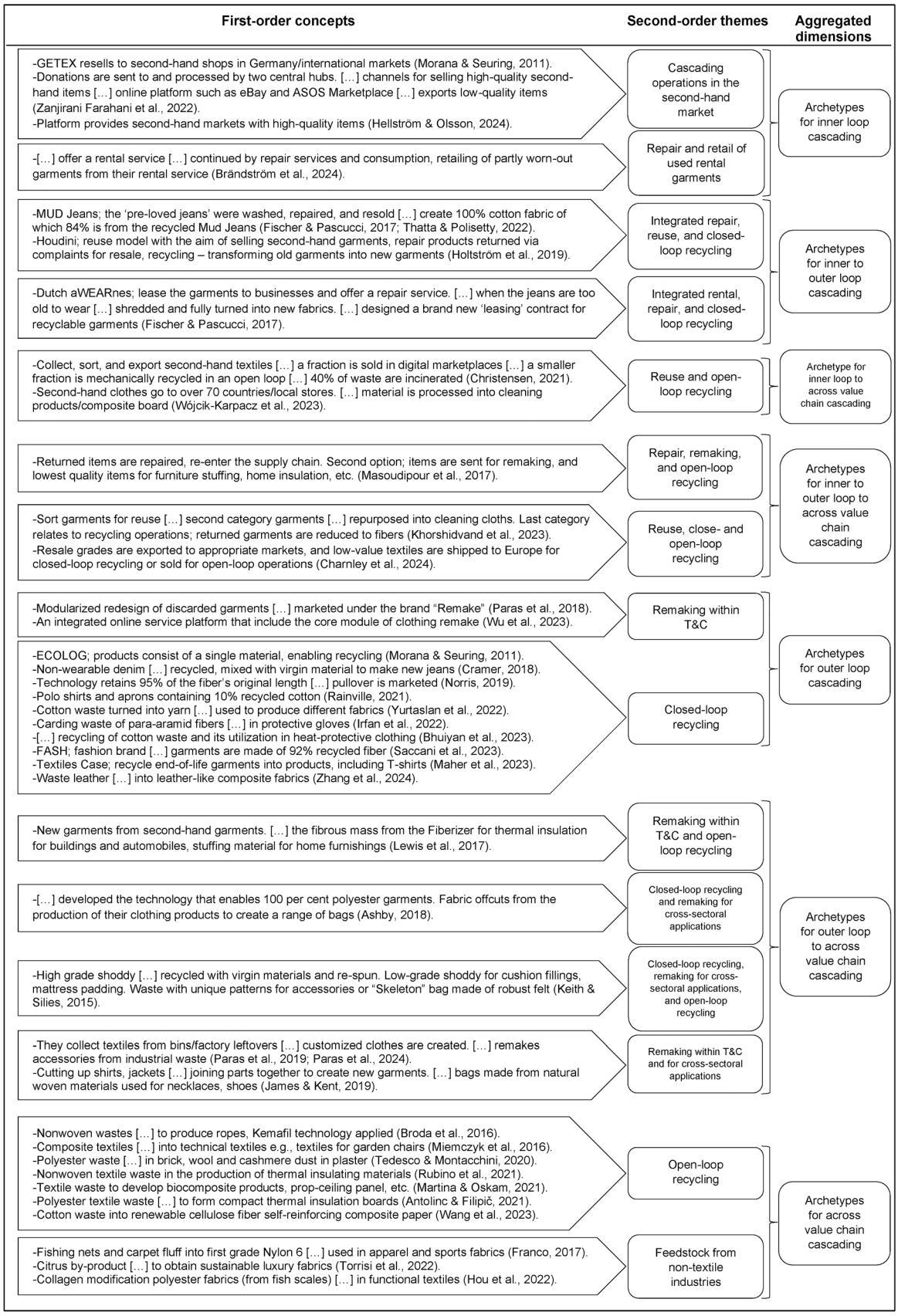


Fig. 3. Data structure.

framework. Where interpretation was needed – particularly for the fourth principle, balancing resource metabolism – greater care was taken. Unlike the first three principles, which were more directly supported by the data, the fourth focused on sustainability and ethical resource management (Sirkin and ten Houten, 1994), and required, for example, calculating product displacement rates – a topic rarely addressed in the literature. Therefore, only the most evident facilitators and barriers were considered under this principle to maintain analytical validity. Once finalized, these were categorized by similarities and differences (see Tables A.2 and A.3 in the Appendix). To ensure inter-coder reliability (Seuring et al., 2012), the remaining authors independently coded a randomly selected subset of the articles using the predefined coding framework. Any differences in interpretation were discussed jointly until consensus was reached. This approach, particularly for complex or subjective principles such as balancing resource metabolism, helped maintain consistency and analytical rigor across coders while enhancing transparency throughout the process.

4. Results

This section presents descriptive statistics of the selected articles and outlines cascading archetypes, along with key facilitators and barriers to cascading operations in the T&C sector.

4.1. Descriptive statistics of selected articles

To provide an overview of the articles included in this study, they were categorized based on four descriptive elements: distribution over years, journal of publication, methods used, and geographical context.

Fig. 4 shows the temporal distribution of publications on cascading operations in the T&C industry, with the earliest article appearing in 2011 – closely aligning with the Ellen MacArthur Foundation's introduction of the CE paradigm in 2010 (Patwa et al., 2021). Despite a few fluctuations, particularly in 2020, the overall trend indicates steady growth, with heightened interest in recent years. Notably, half of the reviewed articles have been published since 2022, reflecting the increasing academic attention to cascading practices.

Furthermore, of the 45 articles reviewed, 44 are full-length and 1 is a conference paper, spanning 31 journals. The Journal of Cleaner Production and the Sustainability journal have the most publications, with 9 and 5 articles, respectively. Two other journals (Resources, Conservation and Recycling as well as Business Strategy and the Environment) each contributed 2 articles, while the remaining journals contributed one each.

Regarding methodological approaches and geographical distribution, Table A.1 in the Appendix provides a comprehensive overview. Case studies overwhelmingly dominate the methodological landscape, appearing in the majority of reviewed articles. This prevalence underscores the exploratory nature of research on cascading practices in the T&C industry, where in-depth contextual analysis is often essential. Beyond traditional case studies, several studies employ experimental approaches based on real-world scalable cases, followed by interview-based research, action research, quantitative analysis, modeling and optimization, and survey-based methods.

The studies also cover a diverse range of geographical contexts, with several examining multinational or collaborative operations. Italy is the most frequently represented country, followed by the Netherlands, Sweden, and the United Kingdom, indicating a strong European focus on cascading operations within the T&C sector. Other European countries also contribute, while some studies extend to other continents, with Asia standing out in particular. This geographical spread highlights both the global relevance of cascading strategies and the concentration of research in specific regions.

4.2. Cascading archetypes in the T&C industry

The SLR identified seven cascading archetypes in the T&C industry, analyzed in Fig. 3 and collectively illustrated in Fig. 5, which visualizes how garments and materials are valorized across circular loops. Inner loops include rental, repair, and reuse at varying quality levels, while outer loops involve remaking and closed-loop recycling. Beyond these, materials are further valorized across value chains through open-loop recycling, non-textile remaking, or energy recovery. The following subdivisions explain each archetype in detail.

4.2.1. Archetypes for inner loop cascading (archetype 1)

Inner-loop cascading archetypes include operations and CBMs such as repair services and innovative reuse/resale of second-hand clothing, focusing on preserving original quality. Literature often refers to this as the collection and sorting of post-consumer clothing, which is redistributed via physical stores, digital marketplaces, and resale platforms. For example, the German company GETEX, emphasizing structured sorting (pre-sorted and washed clothing), cascades high-quality items to domestic and foreign markets for reuse (Morana and Seuring, 2011). In the UK, charities like Oxfam, RTH, and TRAID operate collection and resale systems that valorize used clothing through online, physical, and

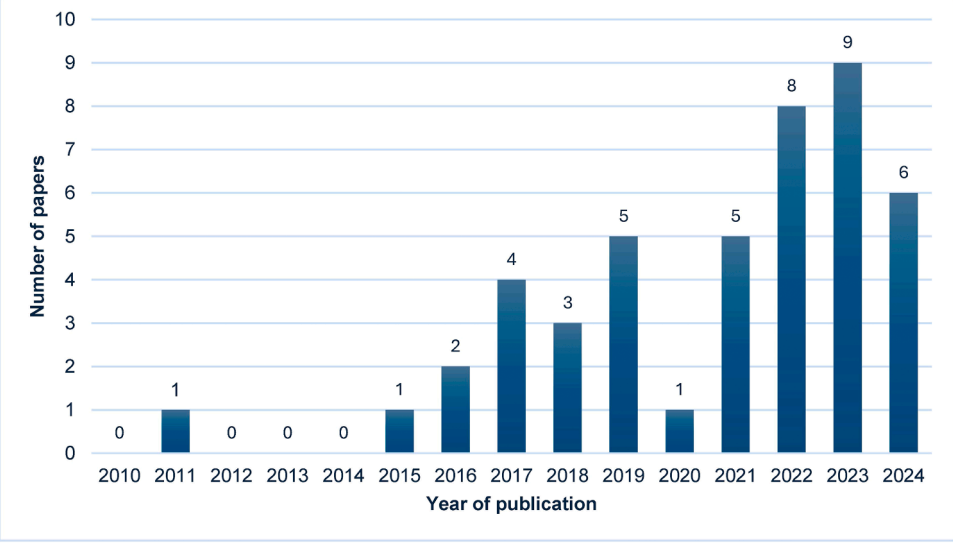


Fig. 4. Distribution of papers by publication year.

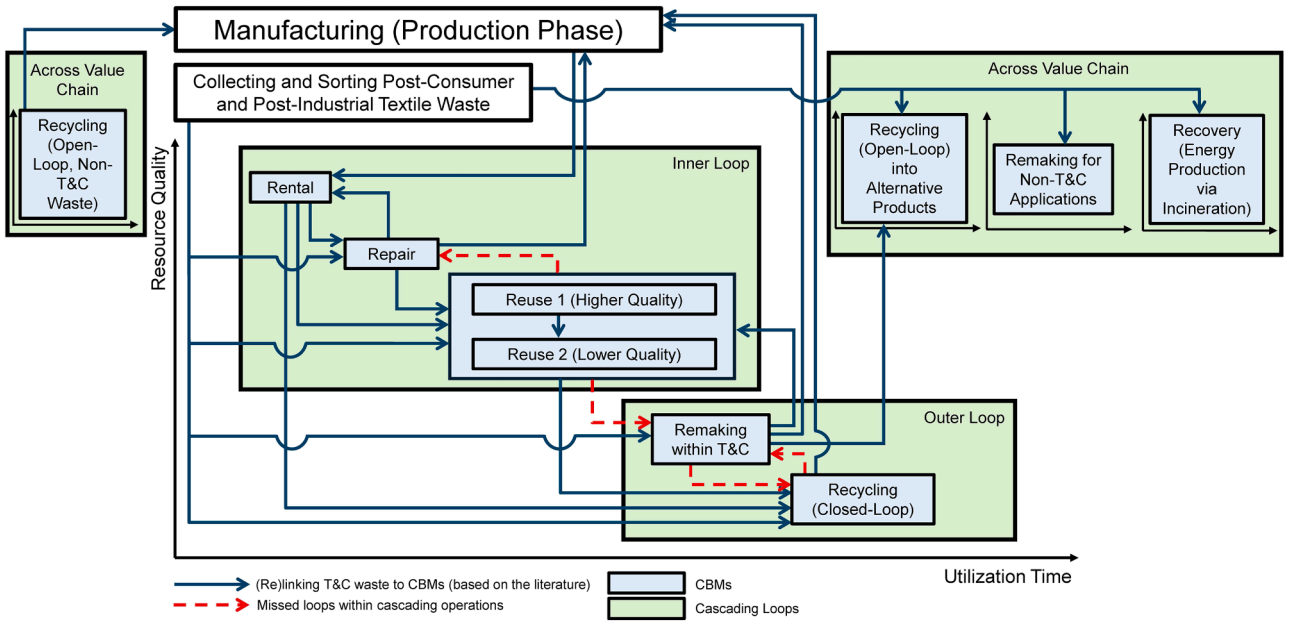


Fig. 5. Aggregated cascading archetypes in the T&C industry.

export channels (Zanjirani Farahani et al., 2022). High-quality items enter primary second-hand markets, while lower-quality or unsold goods are cascaded between shops or exported to low-income countries for reuse. These operations reflect B2B and B2C interactions, enabling cascades within reuse-based CBMs. Another archetype involves platform-based CBMs that scale user-to-user resale opportunities and facilitate access to the second-hand market (Hellström and Olsson, 2024). The literature also highlights an integrated repair-reuse cascade chain archetype, in which a Swedish company repairs worn-out rental outdoor clothing to extend its lifecycle. If rental is not viable, these products are repaired and cascaded into the company's resale platform to prolong their use (Brändström et al., 2024).

4.2.2. *Archetypes for inner to outer loop cascading (archetype 2)*

Inner to outer loop cascading takes a systemic approach by combining inner-loop CBMs with outer-loop processes like recycling, thereby extending product lifespans and closing resource loops. The literature identifies such archetypes in Dutch and Swedish contexts, each applying different CBM combinations. For example, MUD Jeans in the Netherlands operationalizes cascading, driven by repairing garments for reuse (via its Vintage Program) as well as recycling worn-out jeans for cascading materials into new products (Fischer and Pascucci, 2017; Thatta and Polisetty, 2022). Similarly, Houdini's *Re*-project promotes cascading through combining repair, reuse, and recycling CBMs in their operations (Holtström et al., 2019). Dutch aWEARness takes a distinct approach to cascading by embedding the rental model within its system. In this cascading archetype, corporate workwear returned from the rental system is repaired and reintegrated into the rental cycle. When garments no longer meet rental standards, they transition to outer loops for recycling, where materials are transformed into new clothing. These newly produced items are then fed back into the rental system, creating a closed-loop cascade chain aligned with systemic circularity principles (Fischer and Pascucci, 2017).

4.2.3. *Archetype for inner loop to across value chain cascading (archetype 3)*

Operationalizing cascade chains on a global scale often involves valorizing high-quality post-consumer clothing through resale-based CBMs, while prioritizing lower-quality items for open-loop recycling to produce alternative products or for energy recovery. In Denmark, NGOs collect and sort used clothing, where higher-quality items are cascaded to appropriate resale channels, while lower-quality ones are mechanically recycled to cascade materials into the production of alternative products like cleaning cloths, and a significant percentage is incinerated for energy production (Christensen, 2021). Similarly, Poland's VIVE Profit retail chain demonstrates how large-scale operations with integrated logistics and computerized sorting can cascade reusable items to resale channels and reintegrate lower-quality ones into open-loop processes. The company collects, sorts, and cascades quality clothing to over 70 countries and domestic stores for reuse, while recycling non-compliant items into cleaning products or composite boards (Wójcik-Karpacz et al., 2023). These operations resemble a cascade chain archetype combining reuse/resale and open-loop applications across value chains.

4.2.4. *Archetypes for inner to outer loop to across value chain cascading (archetype 4)*

Cascading operations sometimes follow archetypes that integrate inner-loop CBMs with outer-loop strategies, and also extend to open-loop processes by valorizing waste into the production of alternative products. This typically occurs in companies capable of sorting large volumes of used textiles for various CBMs. In an Iranian textile case study, a third party responsible for sorting and (re)distribution plays a key role in cascading products and materials to repair, remaking, and open-loop streams. As noted by Masoudipour et al. (2017), repairable garments are mended and resold as new products at full value, while items unfit for repair – but containing reusable components or materials – are cascaded to a remaking center to become new clothing. In contrast, low-quality items unsuitable for closed-loop operations are cascaded through open-loop recycling, used in the production of furniture stuffing, upholstery, or home insulation. The literature also describes a global cascade supply chain archetype, beginning with textile collection via retail or take-back schemes like H&M. These textiles are then sorted and graded by quality at an international level and cascaded for reuse, closed-loop, and open-loop recycling operations (Charnley et al., 2024; Khorshidvand et al., 2023).

4.2.5. *Archetypes for outer loop cascading (archetype 5)*

In the T&C industry, many companies are specifically leveraging cascading operations to enhance sustainability by focusing on activities

such as remaking or recycling in closed loops. These CBMs operate as sequential outer-loop processes that extend the lifespan of products and materials. The literature on remaking business models highlights the transformation of worn-out garments into reusable clothing through deliberate redesign, complementing a feedback cascading loop from remaking to reuse. Examples of this archetype are seen in China, where community-based services, in collaboration with residents and tailors, provide opportunities to revitalize used clothing (Wu et al., 2023). A Swedish organization has also scaled this digitally, using modular redesign and advanced assembly/disassembly. For instance, a polo T-shirt made from discarded clothes is sold under the brand "Remake" (Paras et al., 2018).

In the recycling archetype, closed-loop operations cascade materials to production cycles. In this process, post-industrial and -consumer waste is cascaded into new fabrics and garments, adding value. Examples include recycling mono-material polyester garments into outdoor clothing, heavy-duty workwear, and safety jackets (Morana and Seuring, 2011). Literature also highlights companies using diverse technologies and collaborations to collect waste, often blending recycled and virgin fibers to make yarn, fabric, towels, pullovers, sweaters, jackets, T-shirts, jeans, and shawls (Maher et al., 2023; Norris, 2019; Saccani et al., 2023; Yurtaslan et al., 2022). The use of recycled fibers in functional garments like protective gloves and heat-protective clothing is another example of sustainable practices (Bhuiyan et al., 2023; Irfan et al., 2022), demonstrating that cascading can serve both mainstream and technical markets. Even leather waste is repurposed into composite yarns for various fabric styles (Zhang et al., 2024). In places like the Netherlands, circular procurement policies support material cascading, enabling new products with recycled content (Cramer, 2018; Rainville, 2021). These recycling-based CBMs show that cascading operations are vital to sustaining material flows in the T&C industry.

4.2.6. *Archetypes for outer loop to across value chain cascading (archetype 6)*

While managing outer-loop CBMs, some companies cascade post-industrial and -consumer waste into processes beyond the T&C industry. These processes, as noted in the literature, follow various archetypes. For instance, a small Haitian business with a "zero waste" approach demonstrates a remaking model by producing new clothes from second-hand garments and cascading leftovers – using a "Fiberizer" – into non-clothing applications such as thermal insulation, stuffing for furnishings and toys, animal bedding, paper goods, and geotextiles (Lewis et al., 2017). In contrast, a UK-based company with a 100 % closed-loop polyester system cascades its production waste into the making of various bags (Ashby, 2018). Another archetype involves cascading operations that sort and reintegrate high-quality materials into closed loops, while relinking lower-quality ones for cushion or mattress filling. Meanwhile, distinctive post-industrial waste is creatively cascaded to make accessories and stylish "Skeleton" bags (Keith and Silies, 2015). A final archetype focuses solely on remaking-based CBMs in both T&C and non-T&C sectors, producing redesigned garments (e.g., shirts, jackets, skirts) from mixed fabrics, and crafting accessories like pencil pouches, wallets, jewelry, headbands, and shoes (James and Kent, 2019; Paras et al., 2019, 2024), promoting sustainability through creative reuse.

4.2.7. *Archetypes for across value chain cascading (archetype 7)*

Cascading can be operationalized as a stand-alone, cross-sector activity by companies seeking innovation. In the literature, this open-loop CBM appears in various forms, such as the T&C industry recycling waste for other sectors, or vice versa. In the first archetype, post-industrial waste like leftover fabrics and fibers is cascaded into the production of thermal and acoustic insulation, composite boards (Antolinc and Filipič, 2021; Rubino et al., 2021), prop ceilings, and modular noise-reducing panels (Martina and Oskam, 2021). Additionally, materials such as polyester have been cascaded into the production of bricks, while wool and cashmere dust have been incorporated into plaster (Tedesco and Montacchini, 2020). Such waste also supports the production of geotextiles (Broda et al., 2016), composite paper (Wang et al., 2023), outdoor furniture, and hose pipes (Miemczyk et al., 2016), enabled by innovative technologies. These examples show that even non-reusable waste can gain value through alternative applications.

In the second archetype, the cascade logic is reversed: raw materials from non-textile industries enter the T&C supply chain. These include processed wastes like citrus by-products (Torrisi et al., 2022), fishing nets, carpet fluff (Franco, 2017), and fish scales (Hou et al., 2022), which are used to produce sports and luxury fabrics as well as functional textiles such as scarves and linings, highlighting the significant potential of cascading waste from other sectors into the T&C industry. For example, brand manufacturers are using polyester modified with collagen extracted from fish scales to improve fabric quality while maintaining its original structure (Hou et al., 2022), thereby contributing to sustainability.

4.3. *Facilitators and barriers to cascading operations in the T&C industry*

This section examines key facilitators and barriers to cascading operations in the T&C sector, using Sirkin and ten Houten's (1994) CCT framework. CCT includes four principles – appropriate fit, augmentation, consecutive relinking, and balancing resource metabolism – originally developed to support sustainable resource use. Table 1 summarizes these factors and their total number of occurrences, along with their effects on inner and outer loops and across value chains, with detailed mappings provided in Appendix Tables A.2 and A.3.

4.3.1. *Appropriate fit*

In the T&C industry, appropriate fit means aligning product and material properties, like appearance, performance, durability, and fiber integrity, with the needs of secondary use. This ensures textiles are repurposed effectively, maximizing utility and minimizing waste. Literature highlights three key facilitators of this principle:

Sorting, cited in 18 studies as shown in Table 1, has emerged as the most frequently mentioned facilitator of appropriate fit and is repeatedly emphasized as the operational gateway to effective reuse and recycling. It plays a pivotal role in matching product/material quality with business needs. This process depends on infrastructure such as sorting centers, collection/distribution stores (Masoudipour et al., 2017; Zanjirani Farahani et al., 2022), computerized lines (Wójcik-Karpacz et al., 2023), and skilled labor (Morana and Seuring, 2007), all working together to direct items into appropriate CBMs based on their quality and characteristics. Its influence extends beyond logistics – sorting acts as a filtering mechanism that determines the fate of materials, shaping whether they enter high-value reuse loops or are downgraded into lower-value recycling streams. Globally, sorting is done manually (often in Asia) or with digital technologies to channel post-consumer textiles into reuse and recycling (Charnley et al., 2024; Khorshidvand et al., 2023). Sometimes, sorting is informed by the careful selection of materials and components to match end-product performance and market demand, supporting effective waste valorization through outer loops (i.e., closed-loop recycling) and across the value chain. This is seen in applications such as heat-protective fabrics, insulation panels, and composite yarns (Bhuiyan et al., 2023; Rubino et al., 2021; Zhang et al., 2024).

While not as dominant as sorting, circular strategies and collaborative models have proven essential in orchestrating the social and organizational dimensions of circularity. These approaches help align products/materials with suitable uses. Inner-loop strategies include MUD Jeans' storytelling to guide second-hand clothing selection (Fischer and Pascucci, 2017) and re-sorting surplus/unsold items for shop-to-shop resale (Zanjirani Farahani et al., 2022). In addition, collaboration between supply chain actors – involving third parties, designers, recyclers, and institutions – facilitates efficient waste routing

Table 1
Facilitators and barriers of cascading operations across loops and principles, with overall occurrences.

Cascading principles	Facilitators	Enabled loops	Overall occurrences	Barriers	Hindered loops	Overall occurrences
Appropriate fit	1-Sorting, material selection, and performance evaluation	Inner loops; Outer loops; Across value chains	18	1-Systemic constraints to effective circular pathways	Inner loops; Outer loops	7
	2-Circular business strategies and collaborative models	Inner loops; Outer loops; Across value chains	7	2-Market fit and demand-side complexities	Outer loops	2
	3-Digital and data-driven systems enabling circular pathways	Inner loops; Outer loops	5			
Augmentation	1-Product life extension strategies	Inner loops; Outer loops; Across value chains	17	1-Misaligned policies and trade restrictions	Inner loops	2
	2-Material life extension strategies	Outer loops; Across value chains	9			
	3-Designing for longevity and circularity	Inner loops; Outer loops	4			
Consecutive Relinking	1-Physical technologies and digital systems supporting circular component and material flows	Outer loops; Across value chains	13	1-System-level constraints and operational complexities in recycling	Outer loops	4
	2-Strategies and mechanisms for product/material reintegration	Inner loops; Outer loops	9	2-Regulatory shortcomings in recycling systems	Outer loops	2
	3-Collaborative circular operations	Inner loops; Outer loops	5			
Balancing Resource Metabolism	1-Strategic corporate and macro-policy interventions for circularity	Outer loops; Across value chains	8	1-Infrastructure gaps and cost inefficiencies in recycling operations	Outer loops	3
	2-Eco-friendly material alternatives	Outer loops; Across value chains	5			
	3-Advanced technologies for sustainable manufacturing	Outer loops; Across value chains	4			

within closed loops and across value chains. For example, Dutch company FurnitureCo partnered with universities to verify material properties and ensure recycled fibers were suitable for non-textile applications (Martina and Oskam, 2021).

Although referenced in only five studies, digital and data-driven systems, in turn, introduce a layer of intelligence that can radically transform the management of cascading operations. Digital platforms can connect high-quality second-hand clothing to broader markets (Hellström and Olsson, 2024) or guide users in selecting sustainable options, such as material types and transportation methods, when choosing between reuse and remaking CBMs (Wu et al., 2023). Their contribution is particularly critical in contexts where manual sorting and decision-making reach their limits. In remaking and recycling, CAD systems (Lewis et al., 2017) and tracking or labeling technologies (Fischer and Pascucci, 2017; Morana and Seuring, 2007) match fabric properties (color, pattern, and design) to specific tasks or ensure that materials are returned to closed-loop systems.

Despite the presence of facilitators, seven studies (Table 1) identify systemic constraints as the main barrier to matching resources with their intended CBMs, undermining cascading operations. On the infrastructure side, limited human resources, inadequate digital and logistical systems, and insufficient capacity for collection and sorting restrict inner-loop reuse (Zanjirani Farahani et al., 2022), often diverting items to less preferred pathways and thereby breaking the cascade chains. In outer loops, especially remake-based CBMs, limited technical expertise and high processing costs push high-quality waste into environmentally suboptimal pathways (Paras et al., 2019). Additional barriers include waste heterogeneity, limited material identification technology, downstream data gaps (Hornbuckle et al., 2023), labor-intensive (pre)sorting, costly processing, and a lack of financial incentives to invest in automated sorting or collecting non-wearable fractions – especially under Extended Producer Responsibility (EPR)-led systems (Boschmeier et al., 2024; Charnley et al., 2024; Reike et al., 2023). These constraints thus are not merely technical – they reflect deeper institutional and policy-level gaps that prevent CBM scaling and integration across regions and actors. These issues compound over time, creating bottlenecks that make closed-loop recycling costly and unattractive for low-value items. In the Netherlands, this has led some firms to focus on recycling post-industrial and -consumer waste into granules or packaging (Reike et al., 2023).

Though less frequently discussed, achieving an appropriate market fit, alongside navigating demand-side complexities, also introduces a subtler but equally critical layer of resistance to cascading operations. The lower quality of recycled materials – often falling short of brand standards – creates a paradox between maintaining brand quality and scaling recycled content (Daddi et al., 2019; Reike et al., 2023). For instance, an Italian producer resells recovered materials instead of reintegrating them into leather-based wearables, because their quality fails to align with the demands of the luxury market (Daddi et al., 2019). This 'escape strategy' reflects a deeper challenge: by prioritizing commercial viability over circular principles, firms reveal how, without shifts in consumer expectations and brand positioning, even technically viable circular solutions may struggle to gain traction.

4.3.2. *Augmentation*

In T&C cascade chains, augmentation refers to processes like repair, redesign, and quality improvement that extend product and material life by enhancing reusability, enabling further cascading without full reprocessing. A literature review on cascading from inner loops to across value chains identified three key facilitators of this principle:

Among the most consistently emphasized enablers of cascading, the strategic extension of product life, highlighted in 17 studies (Table 1), supports the principle of augmentation, which predominantly facilitates inner-loop operations. Companies operationalize rental (Fischer and Pascucci, 2017) and repair services (e.g., Brändström et al., 2024; Holtström et al., 2019) to prolong product lifespan. (*Re*)distribution channels, like physical and online resale (e.g., Zanjirani Farahani et al., 2022), user-to-user platforms (Hellström and Olsson, 2024), and exporting unused/unsold second-hand items (e.g., Morana and Seuring, 2011) to other markets for reuse, also contribute. For instance, UK charity RTH runs a 2-week cull cycle, circulating unsold items between local stores for better utilization (Zanjirani Farahani et al., 2022). These

practices not only delay end-of-life but also redistribute value across geographies and user groups, reinforcing the economic logic of reuse. Similar strategies in outer loops and across value chains include remaking and repurposing textile waste into apparel and non-T&C applications, extending the life of worn-out items before recycling (e.g., Paras et al., 2024).

Closely following product-focused strategies, material life extension has also gained traction as a facilitator, particularly in outer-loop cascades where quality degradation is a key concern. The literature highlights several methods to improve material quality and durability, such as blending virgin and recycled fibers, as practiced by House of Denim in the Netherlands (Cramer, 2018) and Textiles Case in Australia (Maher et al., 2023). Ozone treatment, which removes bacteria and odor, further improves the performance of recycled wool in closed-loop operations (Norris, 2019). Similarly, modifying polyester with fish scale-derived collagen illustrates how augmentation can boost upgradability and fabric performance across value chains (Hou et al., 2022).

Although less frequently discussed, design for longevity and circularity – highlighted in four studies – is key to applying the principle of augmentation in both inner and outer loops. A UK-based company enhances cascading operations through design strategies that prioritize repairability, reusability, and recyclability, even using its own production offcuts for repairs (Ashby, 2018). Similarly, Dutch aWEARness has developed materials that can be recycled multiple times without quality loss, extending their lifespan (Fischer and Pascucci, 2017). These examples show how design decisions made at the beginning of a product's life can ripple through the entire value chain, influencing how easily products can be maintained, reused, or reintegrated.

However, despite these promising developments, progress is often undermined by policy- and trade-related barriers – referenced in only two studies – which exert a disproportionately large impact on inner-loop cascading operations. A key policy barrier is the lack of fiscal incentives and the relatively high taxation of repair services compared to the low cost of virgin raw materials (Brändström et al., 2024). This misalignment creates structural disadvantages for CBMs and subtly reinforces the dominance of linear consumption patterns, making circular alternatives less economically viable. As a result, circularity is promoted in discourse but constrained in practice, discouraging investment and limiting the scalability of repair-reuse cascade chains. A similar contradiction exists in global trade, where several African countries, major recipients of European second-hand clothing, impose high tariffs on second-hand clothing imports, while allowing imports of new clothing from China without tariffs (Charnley et al., 2024). Such asymmetries not only distort market signals but also shape the global geography of textile flows, often to the detriment of circularity.

4.3.3. *Consecutive relinking*

In the T&C sector, consecutive relinking refers to the smooth transition of used textiles through reverse supply chain stages, from inner- to outer-loop CBMs like reuse to recycling. Its success depends on strong infrastructure, collaboration, and design strategies that enable seamless reintegration into new uses. Literature identifies three main facilitators of this transition within and between inner and outer loops and across value chains:

Technological innovation, particularly in physical and digital systems, has consistently emerged as a key enabler of consecutive relinking, forming the technical backbone of cascading operations, highlighted in 13 studies (Table 1). These innovations range from closed-loop recycling of wool and polyester (e.g., Ashby, 2018; Norris, 2019) to depolymerizing fishing nets and carpet fluff for garments (Franco, 2017). Moreover, Kemafil technology enables coarse rope production from recycled fibers (Broda et al., 2016), while denim waste is converted into composite paper through wet-laid and hot-pressing techniques (Wang et al., 2023). Digital tools like CAD (Lewis et al., 2017) and redesign algorithms (Paras et al., 2018) also support remaking and reintegrating worn garments into reuse-based CBMs. These technologies do more than just enable remaking or recycling – they reconfigure T&C resource flows by allowing previously discarded garments to re-enter value chains with renewed utility, thereby facilitating outer-loop operations and enabling consecutive relinking across the value chain.

Identified in nine studies, strategic mechanisms for product and material relinking follow technological enablers and play a vital role in operationalizing cascading across loops. At the company level, repair not only enables augmentation but also strengthens inner-loop cascades by consecutively relinking worn clothing to reuse, as seen with MUD Jeans, Dutch aWEARness, and Houdini Sportswear (Fischer and Pascucci, 2017; Holtström et al., 2019; Thatta and Polisetty, 2022). MUD Jeans and Dutch aWEARness also retain product ownership, facilitating system-wide, cascaded CBMs like rental, repair, reuse, and recycling. Some companies design for recyclability to support closed-loop operations (Maher et al., 2023), while others, such as Hoooked in Portugal, reintegrate textile waste like edge trims into the value chain through digital platforms (Schmutz and Som, 2022). At the policy level, measures like mandatory post-consumer waste collection and sustainable public procurement bolster closed-loop recycling (Christensen, 2021; Rainville, 2021). A notable case is the Dutch Ministry of Defense's procurement policy, which uses a rental and recycling model for protective clothing, also adopted by the national police and fire brigade (Cramer, 2018).

Although less frequently emphasized, collaborative circular operations are also essential for bridging gaps between actors and enabling system-wide relinking. For example, Dutch aWEARness facilitates cascading from inner to outer loops (rental, repair, recycling) by orchestrating the value chain and connecting system actors (Fischer and Pascucci, 2017). Similarly, the Textiles Case partners with stakeholders in waste collection, decommissioning, shredding, ragging, and spinning to leverage existing expertise and infrastructure for a domestic closed-loop recycling operation (Maher et al., 2023). These collaborations are not just logistical – they represent a form of distributed governance, where knowledge, resources, and responsibilities are shared to overcome fragmentation. Additionally, intermediaries play a key role by hosting workshops, offering consultancy, and identifying market opportunities. They engage suppliers and recyclers to assess feasibility, costs, and technical needs for closed-loop operations (Rainville, 2021).

Despite these facilitators, system-level constraints, noted in four studies, remain a major barrier to effective relinking and are often cited as bottlenecks to scaling cascading operations. A key issue is inadequate infrastructure and industrial-scale technology. For instance, in Switzerland in 2019, insufficient facilities resulted in over half of post-industrial textile waste being incinerated or diverted to open-loop recycling (Schmutz and Som, 2022). Even when recycling is technically viable, the absence of organized collection systems for supplying materials, as seen with ECOLOG in Germany, impedes closed-loop cascading operations (Morana and Seuring, 2011). Irregular and low-volume waste further reduces economic appeal for recyclers (Maher et al., 2023; Schmutz and Som, 2022). These limitations reflect a mismatch between circular ambitions and the operational realities of waste management, where scale, consistency, and coordination are often lacking. Moreover, garment complexity, including zippers and buttons, complicates disassembly and recyclability (Maher et al., 2023). Outsourcing production further weakens companies' control over recycling and relinking their own waste, as observed among some Finnish apparel manufacturers (Salmi and Kaipia, 2022).

Regulatory shortcomings, noted in only two studies, also pose significant barriers to outer-loop cascading in textile recycling. The absence of harmonized regulations for hazardous substances and the ambiguity surrounding standards for recycled inputs contribute to uncertainty, particularly in cross-border contexts (Brändström et al., 2024). In Italy, more stringent hygiene requirements for products containing recycled wool, compared to international norms, further complicate material relinking efforts. While this tension between environmental and

health-related objectives is partially mitigated through defensive strategies – such as rigorous sorting of recycled raw materials and ensuring traceability to foster customer trust (Daddi et al., 2019) – these measures remain insufficient. The persistence of such regulatory tensions underscores the urgent need for integrated policy frameworks that reconcile safety and sustainability, rather than allowing one to compromise the other.

4.3.4. Balancing resource metabolism

In the T&C sector, balancing resource metabolism involves designing and managing material flows to reduce waste and promote reuse and recycling. This demands system-level coordination over isolated efforts. Literature identifies three key facilitators for this, especially in outer loops and across value chain operations:

Strategic corporate actions and macro-level policies, identified in eight studies (Table 1), have emerged as the most prominent drivers in reducing dependence on virgin resources, shaping both production logic and regulatory frameworks. At the corporate level, fully utilizing post-consumer waste (e.g., Lewis et al., 2017) or designing all-natural, biodegradable products (Maher et al., 2023) help to achieve CE. The start-up FASH also employs a pre-sale model and no-discount policy to align production with demand, minimize inventory, and aid supplier planning (Saccani et al., 2023). These interventions reflect a shift from reactive waste management to proactive resource stewardship, where firms take responsibility for material flows from design to post-use. At the policy level, EU initiatives promoting high-quality recyclable materials (Fischer and Pascucci, 2017) and circular procurement mandating a minimum recycled content in clothing (Cramer, 2018; Rainville, 2021) reduce dependence on raw materials. Such policies signal a growing institutional commitment to embedding circularity into procurement and production standards.

Eco-friendly material alternatives, noted in five studies, provide another critical leverage point for transforming resource metabolism and reducing environmental impacts. Literature highlights the replacement of traditional materials with sustainable substitutes that lower energy consumption, as shown in life cycle assessments. For instance, wool-based insulation that includes 41 % phase change materials reduces annual energy consumption by 26 % for air conditioning and by 31.6 % for cooling, compared to conventional polyurethane foam (Rubino et al., 2021). Biodegradable materials, like yarn made from citrus by-products (Torrisi et al., 2022) or leather-like waste (Zhang et al., 2024) also contribute to sustainability initiatives. These innovations not only reduce ecological footprints but also challenge the dominance of petrochemical-based inputs, offering regenerative alternatives that align with circular principles.

Closely following eco-friendly material alternatives, advanced technologies for sustainable manufacturing, noted in four studies, play a transformative role in enabling closed-loop cascading systems. For example, RecycleCo in Italy uses depolymerization technology to produce 100 % recycled synthetic yarn from non-woven waste, allowing for unlimited recycling while maintaining quality (Franco, 2017). Closed-loop advanced technologies have also enabled full polyester recycling into new garments (Ashby, 2018; Fischer and Pascucci, 2017). Furthermore, digital technologies in large-scale garment remaking reduce resource input and waste more effectively than ad hoc methods, aligning with ecological capacities and enabling industry-wide impact (Paras et al., 2018). These technologies represent a frontier in circular innovation, where material cascading is no longer constrained by degradation or open-loop recycling.

Despite these advancements, infrastructure gaps and cost inefficiencies, noted in three studies (Table 1), remain persistent barriers to balancing resource metabolism. In Australia, for example, the lack of local spinning facilities compels companies to send fibers abroad for processing, increasing emissions and reducing environmental benefits (Maher et al., 2023), thereby undermining progress toward circularity. Similar rebound effects occur in Sweden, where recycled fabrics are imported from Japan due to limited domestic supply. Yet the sheer distance to such recycling centers not only counteracts the intended environmental gains but also inflates consumer prices (Brändström et al., 2024), raising questions about the overall viability and equity of scaling recycled textiles in global markets. In Switzerland, high logistics costs make recycling more expensive than incineration (Schmutz and Som, 2022), breaking closed-loop cascade chains and disrupting resource metabolism.

Fig. 6, illustrating CBMs operating within the T&C sector, summarizes the analysis of facilitators and barriers across the four cascading principles.

5. Discussion

This paper explores how cascading is operationalized across CBMs and loops in the T&C industry. To understand how cascading loops function in this sector, the study synthesizes findings from seven archetypes and four CCT principles – appropriate fit, augmentation, consecutive relinking, and balancing resource metabolism. Originally proposed by Sirkin and ten Houten (1994), these principles offer a systemic lens for evaluating CBMs beyond isolated reuse or recycling efforts. Recent literature (e.g., Bocken et al., 2016; Charnley et al., 2024; Lüdeke-Freund et al., 2018; Reike et al., 2023) has expanded on these principles, emphasizing the need for coordinated, multi-loop strategies. Drawing from this body of work and empirical findings, we identify four critical dimensions underpinning cascading operations: (1) the logic behind firms' choices, (2) structural, technological, and operational synergies, (3) value chain coordination and design for cascading loops, and (4) the challenges of consecutive relinking and the need for stronger collaboration.

5.1. The logic behind firms' choices

Grounded in the findings, companies do not follow a rigid, hierarchical, or sequential model for operationalizing CBMs; rather, they selectively adopt specific archetypes based on product and material characteristics, available infrastructure, and market conditions. While this reflects the principle of appropriate fit, where sorting and material matching facilitate the feasibility of reuse and recycling (Masoudipour et al., 2017; Zanjirani Farahani et al., 2022), loops such as non-T&C remaking and open-loop recycling are often prioritized (e.g., archetypes 3 and 6) – not due to their higher value retention, but rather because they are commercially viable. This prioritization tends to sideline inner-loop CBMs such as repair, which focus on value augmentation; as shown in the results (e.g., archetypes including inner loops), repair operations are inconsistently prioritized, especially in European contexts such as the UK, Germany, and Poland (Christensen, 2021; Morana and Seuring, 2011; Wójcik-Karpacz et al., 2023), aligning with Dehghannejad et al. (2025), who found that in the European context, repair operations are frequently deprioritized due to their high costs and limited value retention, particularly when compared to the affordability of new clothing. These choices are further shaped by system-level and operational barriers, including the absence of standardized sorting, fragmented reverse logistics, and technological constraints in recycling, all of which hinder the scalability of higher-value loops like reuse and closed-loop recycling operations. Additionally, policy and regulatory frameworks, coupled with market disincentives, reinforce these suboptimal open-loop pathways, leading to a landscape where cascading is driven by what is practically achievable within existing systems. This supports the argument by Bocken et al. (2016) that CBMs are governed by both technical feasibility and economic viability, often overriding considerations such as consumption rate and salvageability – two key dimensions of the CCT. Based on these findings, the logic behind firms' choices reveals that *the T&C industry's current cascading operations stem largely from pragmatic considerations rather than a commitment to normative circular principles.*

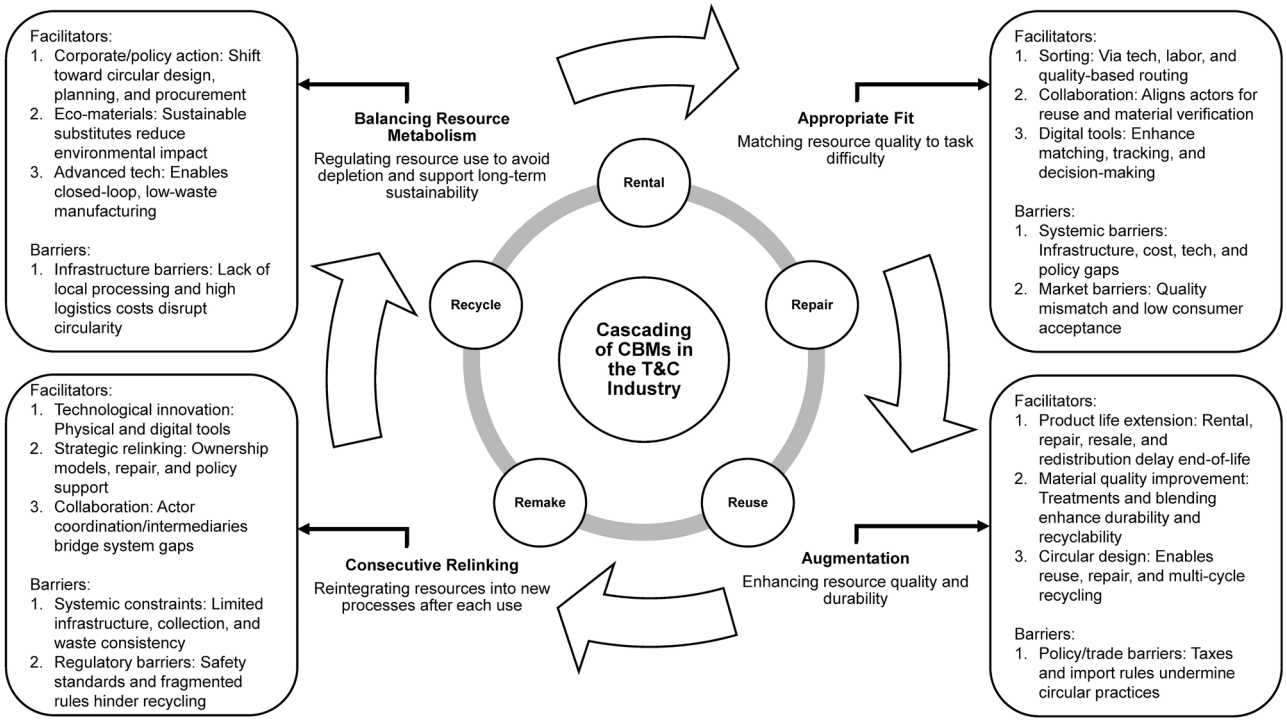


Fig. 6. Facilitators and barriers to cascading principles in the T&C industry.

5.2. Structural, technological, and operational synergies

Findings also indicate that several companies are increasingly adopting CBMs to extract value from used textiles, aligning with cascading principles. This approach is closely tied to augmentation and appropriate fit, where sorting, repair, and material quality improvements enable garments to circulate through high-value loops. For instance, repair facilitates multiple uses of a single garment, allowing flow between rental and reuse cascades and bridging augmentation with consecutive relinking. As illustrated in Archetypes 2 and 4, companies such as Houdini Sportswear and MUD Jeans (Holtström et al., 2019; Thatta and Polisetty, 2022) integrate repair and recycling to extend product lifespans and close material loops. Relinking is further supported by collaborations, policy initiatives, and advanced technologies, with digital tools playing an increasingly central role in coordinating and optimizing these processes. These practices demonstrate how CBMs operationalize the technical cycle of the CE by preserving the material, energy, and labor embedded in products – slowing resource flows in inner loops and closing them in outer loops (Brenner and Drdla, 2023; Hopkinson et al., 2018). Yet despite these advances, Europe continues to face systemic barriers: inner-loop reuse is constrained by limited human resources, inadequate logistics, and insufficient sorting capacity (Zanjirani Farahani et al., 2022), while outer-loop CBMs such as recycling are hindered by costly processing, labor-intensive sorting, and weak financial incentives under EPR schemes (Boschmeier et al., 2024; Charnley et al., 2024; Reike et al., 2023). These barriers are reflected in the Archetypes, where open-loop recycling and cross-sector applications often emerge as fallback strategies due to insufficient infrastructure and policy support for closed-loop cascading operations. Addressing these constraints, as emphasized by the European Environment Agency (2023), requires a coordinated effort to prioritize less energy-intensive inner loops over outer-loop CBMs. To achieve this, *cascade chains must be operationalized in ways that foster structural, technological, and operational synergies – translating cascading principles into circular systems that are cohesive, collaborative, and value-driven.*

5.3. Value chain coordination and design for cascading loops

As illustrated in Fig. 5, inner-loop strategies, such as rental, repair, and resale-based CBMs, alongside closed-loop recycling and material reintegration, collectively contribute to a robust cascade chain that retains product and material value across the T&C industry. However, secondary cascade chains (i.e., cascading across value chains) have become a mainstream operation (Archetypes 3, 4, 6, and 7), especially when fiber-to-fiber solutions are limited. This type of cascading does not inherently complement the loop strategies emphasized by Bocken et al. (2016), as it may fail to integrate slowing and closing mechanisms effectively, thereby weakening systemic circularity. This limitation underscores the importance of balancing resource metabolism, a principle that emphasizes coordinated design and procurement strategies to regulate material flows and reduce leakage throughout the value chain (Sirkin and ten Houten, 1994). To unlock future value, cascading must be redefined as a traceable and well-orchestrated system in which products are intentionally designed to pass through multiple value-retention stages, maximizing both longevity and resource efficiency. This systemic approach is exemplified in Archetype 2, where companies such as Dutch aWEARness and Houdini Sportswear integrate rental, repair, reuse, and closed-loop recycling through advanced technologies and strategies such as design for repairability, trace-and-track systems, value chain coordination, and product ownership retention (Fischer and Pascucci, 2017; Holtström et al., 2019). These mechanisms demonstrate how structured interventions can preserve material integrity, support responsible resource management, and advance the circular economy's broader goal of regenerating natural systems (Ellen MacArthur Foundation, 2025). Therefore, *value chain coordination in the T&C sector emerges not merely as an operational concern but as a strategic imperative, shaping how cascading loops are designed, operationalized, and sustained across sectors.*

5.4. Challenges of consecutive relinking and the need for stronger collaboration

Despite recent progress, significant circular value in the T&C reverse

supply chain remains untapped (see, e.g., red dotted lines in Fig. 5). Repair often takes a backseat to reuse, rather than being integrated into a circular system that completes feedback loops and recirculates products within inner loops. Zanjirani Farahani et al. (2022) attribute this to the low resale value of second-hand garments, which weakens the economic case for repair. Moreover, transitions from inner to outer loops remain inefficient, leading to lost value. This inefficiency directly reflects the principle of consecutive relinking, where the lack of infrastructure, collaboration, and feedback systems hinders smooth transitions between CBMs. For instance, while remaking can reconnect outer loops with reuse-oriented CBMs, it lacks the feedback systems needed for scaling due to technical barriers, high costs, and weak market demand (Paras et al., 2019). Additionally, the absence of direct links between remaking and closed-loop recycling results in further uncaptured value. These gaps, especially in outer-loop operations, reveal that CBMs often function in isolation rather than as part of an integrated system. This fragmentation is evident in Archetypes 5 and 6, where remaking and recycling operate independently, limiting systemic integration. Such fragmentation underscores the need for a comprehensive CE-cascading framework that aligns design, infrastructure, and policy to support consecutive relinking and long-term resource stewardship (Campbell-Johnston et al., 2020). In this regard, Charnley et al. (2024) and Schmutz and Som (2022) call for stronger collaboration and centralized operations. For example, regeneration zones that integrate sorting, repair, and fiber-to-fiber recycling could streamline logistics, scale circular practices, and retain textiles within the system. These collaborative approaches not only improve the fit between resource quality and task demand but also, as Sirkin and ten Houten (1994) stress, help operationalize the principle of consecutive relinking by turning the otherwise separate principles of appropriate fit and augmentation into a cohesive cascade chain. Therefore, *enabling consecutive relinking through system-wide collaboration and integration is essential to overcoming fragmentation and realizing the full circular potential of cascading in the T&C industry*.

However, these insights should be interpreted in light of the study's limitations, which also point to directions for future research. First, although this review explored the operationalization of cascading across CBMs, many included studies only briefly addressed related concepts without examining the mechanisms and dynamics of cascading operations. This highlights the need for research to unpack and articulate cascading practices more explicitly, both conceptually and empirically. The temporal distribution of publications – beginning in 2011 and accelerating post-2022 – suggests that cascading is a relatively recent and evolving research area. This recency implies that many studies remain exploratory, lacking longitudinal data or robust theoretical grounding. Case study methodologies dominate, offering contextual depth but limiting generalizability; future work should employ longitudinal and comparative designs to trace cascading strategies over time, particularly in relation to shifting policy landscapes and market dynamics. Second, the concentration of studies in Europe reveals a regional bias, leaving gaps in understanding cascading practices elsewhere, especially in the Global South. Future research should explore underrepresented contexts, including emerging economies where textile production and waste generation are significant but underexamined. Third, several identified barriers appeared contradictory, revealing paradoxical tensions within cascading strategies. While firms often adopt ad hoc solutions to address them, the limited scope of our study restricts deeper analysis. Given their potential to disrupt systemic circularity efforts, future research should investigate how such tensions manifest along cascading principles and how they can be managed through organizational capabilities, resource mobilization, and proactive policy frameworks. Finally, contradictions may arise when operationalizing cascade chains – for example, extending material lifespan through blending or augmentation may compromise recyclability and hinder material relinking due to the immaturity of technologies for breaking blends in cascading processes. These trade-offs point to theoretical gaps and opportunities for theory development, adding nuance and realism to strategies for circular transitions in the T&C industry.

6. Conclusion

This study conducted an SLR to explore how cascading is operationalized in the T&C industry and to identify underlying facilitators and barriers. Seven distinct cascading archetypes were identified, reflecting the diversity of CBMs in practice. While the field remains emergent, several enabling factors, such as efficient sorting, advanced technologies, circular design strategies, life extension techniques, supply chain collaboration and coordination, and supportive policies, were found to support cascading operations. Conversely, infrastructural gaps, market and safety constraints, limited local processing capacity, and regulatory and trade restrictions continue to limit the scalability and integration of CBMs. By mapping these dynamics across the four cascading principles, the study reveals how micro-level actions (e.g., repair, sorting, reuse) are linked to broader goals of material efficiency and systemic circularity. Realizing the full potential of cascading requires structural, technological, and operational synergies, effective value chain coordination, and stronger collaboration.

This study offers both theoretical and practical contributions. Theoretically, it advances the application of CCT within the T&C industry by identifying how real-world practices deviate from ideal cascade chains. Practically, it provides actionable guidance for businesses and policymakers. A coordinated approach to CBMs that acknowledges the interdependence of cascading of loops can enhance product value, reduce waste, and improve resource efficiency. Companies that treat cascading as a systemic process and understand loop dynamics are better positioned to manage resources, meet sustainability targets, and build long-term resilience. Identifying key facilitators and barriers also helps businesses valorize textile waste and supports policymakers in designing targeted, context-sensitive interventions. To translate these insights into action, companies can invest in traceability systems, modular design, and collaborative infrastructures such as regeneration zones. Policymakers, in turn, can incentivize inner-loop operations and support digital and logistical integration to scale circular practices across the T&C value chain.

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CRediT authorship contribution statement

Mohammadreza Dehghannejad: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Rudrajeet Pal:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Kanchana Dissanayake:** Writing – review & editing, Validation, Supervision, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Mohammadreza Dehghannejad reports financial support was provided by Swedish Research Council Formas. Rudrajeet Pal reports financial support was provided by Swedish Research Council Formas. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A1
Table A2
Table A3

Table A1
Mapping of articles according to context, methods, cascading principles, facilitators, and barriers.

#	Reference	Geographical Context	Method	Appropriate Fit	Augmentation	Consecutive Relinking	Balancing Resource Metabolism	Facilitator	Barrier
1	Morana and Seuring (2011)	Germany	Case study	✓	✓	✓		✓	✓
2	Keith and Silies (2015)	Scotland	Case study	✓	✓	✓		✓	
3	Broda et al. (2016)	Poland	Experimental study (scalable real-world case)	✓	✓	✓		✓	
4	Miemczyk et al. (2016)	Italy-France	Case study			✓		✓	
5	Franco (2017)	Italy	Case study	✓		✓	✓	✓	
6	Fischer and Pascucci (2017)	The Netherlands	Case study	✓	✓	✓	✓	✓	
7	Masoudipour et al. (2017)	Iran	Modeling and optimization	✓	✓	✓	✓	✓	
8	Lewis et al. (2017)	United States-Haiti	Case study with action research approach	✓	✓	✓	✓	✓	
9	Ashby (2018)	United Kingdom	Case study	✓	✓	✓	✓	✓	
10	Cramer (2018)	The Netherlands	Case study	✓	✓	✓	✓	✓	
11	Paras et al. (2018)	Sweden	Case-based modeling study	✓	✓	✓	✓	✓	
12	James and Kent (2019)	Ghana	Action research	✓	✓	✓		✓	
13	Paras et al. (2019)	Romania	Case study and quantitative analysis	✓	✓	✓		✓	✓
14	Norris (2019)	Spain	Case study		✓	✓		✓	
15	Daddi et al. (2019)	Italy	Case study						✓
16	Holtström et al. (2019)	Sweden	Case study	✓	✓	✓		✓	
17	Tedesco and Montacchini (2020)	Italy	Experimental study (scalable real-world case)	✓	✓	✓	✓	✓	
18	Christensen (2021)	Denmark	Action research	✓	✓	✓		✓	
19	Rainville (2021)	The Netherlands	Case study			✓	✓	✓	
20	Rubino et al. (2021)	Italy	Experimental study (scalable real-world case)	✓	✓	✓	✓	✓	
21	Martina and Oskam (2021)	The Netherlands	Case study	✓	✓	✓		✓	
22	Antolinc and Filipič (2021)	Slovenia	Experimental study (scalable real-world case)	✓		✓		✓	
23	Schmutz and Som (2022)	Switzerland	Survey					✓	✓
24	Yurtaslan et al. (2022)	Türkiye	Experimental study (scalable real-world case)	✓	✓	✓	✓	✓	
25	Torrisi et al. (2022)	Italy	Case study	✓	✓	✓	✓	✓	
26	Hou et al. (2022)	Taiwan	Experimental study (scalable real-world case)	✓	✓	✓	✓	✓	
27	Thatta and Polisetty (2022)	The Netherlands	Case study	✓	✓	✓		✓	
28	Zanjirani Farahani et al. (2022)	United Kingdom	Interview-based study	✓	✓	✓		✓	✓
29	Irfan et al. (2022)	South Korea-Pakistan	Experimental study (scalable real-world case)	✓	✓	✓		✓	
30	Salmi and Kaipia (2022)	Finland	Case study						✓
31	Khorshidvand et al. (2023)	Australia	Case study and quantitative analysis	✓	✓	✓		✓	

(continued on next page)

Table A1 (*continued*)

#	Reference	Geographical Context	Method	Appropriate Fit	Augmentation	Consecutive Relinking	Balancing Resource Metabolism	Facilitator	Barrier
32	Wu et al. (2023)	China	Case study	✓	✓	✓		✓	
33	Hornbuckle et al. (2023)	United Kingdom	Interview-based study						✓
34	Wang et al. (2023)	China	Experimental study (scalable real-world case)	✓		✓		✓	
35	Reike et al. (2023)	The Netherlands	Case study						✓
36	Wójcik-Karpacz et al. (2023)	Poland	Case study	✓	✓	✓		✓	
37	Bhuiyan et al. (2023)	Bangladesh-China	Experimental study (scalable real-world case)	✓	✓	✓		✓	
38	Saccani et al. (2023)	Italy	Case study			✓	✓	✓	
39	Maher et al. (2023)	Australia	Case study	✓	✓	✓	✓	✓	✓
40	Paras et al. (2024)	Romania	Interview-based study	✓	✓	✓		✓	
41	Hellström and Olsson (2024)	Sweden	Interview-based study	✓	✓	✓		✓	
42	Zhang et al. (2024)	China	Experimental study (scalable real-world case)	✓	✓	✓	✓	✓	
43	Boschmeier et al. (2024)	EU	Survey						✓
44	Charnley et al. (2024)	Across European countries	Interview-based study	✓		✓		✓	✓
45	Brändström et al. (2024)	Sweden	Case study		✓	✓		✓	✓

Table A2
Facilitators of cascading operations aligned with cascading principles, loops enabled, and literature frequency.

Cascading principles	Facilitators	How cascading is supported	Enabled loops	Occurrences in literature	Supporting literature
Appropriate Fit	1-Sorting, material selection, and performance evaluation	Sorting across multiple loops enabled by infrastructure, technologies, and skills	Inner loops; Outer loops; Across value chains	8	Charnley et al. (2024); Fischer and Pascucci (2017); Keith and Silies (2015); Khorshidvand et al. (2023); Masoudipour et al. (2017); Morana and Seuring (2011); Wójcik-Karpacz et al. (2023); Zanjirani Farahani et al. (2022)
		Material selection and performance evaluation – valorization of waste for both closed- and open-loop applications	Outer loops; Across value chains	10	Bhuiyan et al. (2023); Hou et al. (2022); Irfan et al. (2022); Martina and Oskam (2021); Paras et al. (2024); Paras et al. (2018); Rubino et al. (2021); Tedesco and Montacchini (2020); Torrisi et al. (2022); Zhang et al. (2024)
	2-Circular business strategies and collaborative models	Storytelling; shop-to-shop (cull cycle) resale	Inner loops	2	Fischer and Pascucci (2017); Zanjirani Farahani et al. (2022)
		Various collaborative approaches (e.g., involving third parties, designers, recyclers, and institutions)	Inner loops; Outer loops; Across value chains	5	Cramer (2018); Franco (2017); Martina and Oskam (2021); Masoudipour et al. (2017); Zanjirani Farahani et al. (2022)
	3-Digital and data-driven systems enabling circular pathways	Second-hand market platform; sustainability assessment platform	Inner loops; Outer loops	2	Hellström and Olsson (2024); Wu et al. (2023)
		CAD systems for color, pattern, and design matching in remaking	Outer loops	1	Lewis et al. (2017)
		Labeling and track-and-trace systems supporting closed-loop recycling and material compatibility	Outer loops	2	Fischer and Pascucci (2017); Morana and Seuring (2011)
Augmentation	1-Product life extension strategies	Rental, repair, resale (through various channels), and remaking practices	Inner loops; Outer loops; Across value chains	17	Ashby (2018); Brändström et al. (2024); Christensen (2021); Fischer and Pascucci (2017); Hellström and Olsson (2024); Holtström et al. (2019); James and Kent (2019); Keith and Silies (2015); Lewis et al. (2017); Masoudipour et al. (2017); Morana and Seuring (2011); Paras et al. (2019); Paras et al. (2024); Paras et al. (2018); Thatta and Polisetty (2022); Wójcik-Karpacz et al. (2023); Zanjirani Farahani et al. (2022)

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Table A2 (*continued*)

Cascading principles	Facilitators	How cascading is supported	Enabled loops	Occurrences in literature	Supporting literature
	2-Material life extension strategies	Improving quality through material modification, treatment, and blending	Outer loops; Across value chains	9	Cramer (2018); Hou et al. (2022); Irfan et al. (2022); Keith and Silies (2015); Maher et al. (2023); Norris (2019); Rubino et al. (2021); Yurtaslan et al. (2022); Zhang et al. (2024)
	3-Designing for longevity and circularity	Design for repairability, reusability, and recyclability	Inner loops; Outer loops	4	Ashby (2018); Brändström et al. (2024); Fischer and Pascucci (2017); Holtström et al. (2019)
Consecutive Relinking	1-Physical technologies and digital systems supporting circular component and material flows	Advanced recycling technologies and techniques – enabling material reintegration	Outer loops; Across value chains	11	Ashby (2018); Broda et al. (2016); Fischer and Pascucci (2017); Franco (2017); Lewis et al. (2017); Martina and Oskam (2021); Miemczyk et al. (2016); Norris (2019); Wang et al. (2023); Wójcik-Karpacz et al. (2023); Zhang et al. (2024)
		Digital systems – CAD and group genetic algorithm	Outer loops	2	Lewis et al. (2017); Paras et al. (2018)
	2-Strategies and mechanisms for product/ material reintegration	Relinking strategies – implementing repair models; retaining product/ material ownership; designing for recyclability; reselling leftover fabrics online	Inner loops; Outer loops	6	Brändström et al. (2024); Fischer and Pascucci (2017); Holtström et al. (2019); Maher et al. (2023); Schmutz and Som (2022); Thatta and Polisetty (2022)
		Legally enforced collection systems and procurement policies supporting material reintegration	Outer loops	3	Christensen (2021); Cramer (2018); Rainville (2021)
	3-Collaborative circular operations	Value chain collaboration, intermediation, and orchestration	Inner loops; Outer loops	5	Charnley et al. (2024); Fischer and Pascucci (2017); Maher et al. (2023); Rainville (2021); Zanjirani Farahani et al. (2022)
Balancing Resource Metabolism	1-Strategic corporate and macro-policy interventions for circularity	Corporate strategies – zero-waste approach; design for recycling or decomposition; pre-sale, non-discount pricing, and demand-aligned production	Outer loops; Across value chains	5	Lewis et al. (2017); Maher et al. (2023); Masoudipour et al. (2017); Saccani et al. (2023); Yurtaslan et al. (2022)
		EU funding and circular procurement policies	Outer loops	3	Cramer (2018); Fischer and Pascucci (2017); Rainville (2021)
	2-Eco-friendly material alternatives	Sustainable material substitution – materials that reduce ecological impact throughout the product life cycle	Outer loops; Across value chains	5	Hou et al. (2022); Rubino et al. (2021); Tedesco and Montacchini (2020); Torrisi et al. (2022); Zhang et al. (2024)
	3-Advanced technologies for sustainable manufacturing	Advanced modularized remaking processes and closed-loop technologies enabling 100 % material regeneration	Outer loops; Across value chains	4	Ashby (2018); Fischer and Pascucci (2017); Franco (2017); Paras et al. (2018)

Table A3
Barriers of cascading operations aligned with cascading principles, loops hindered, and literature frequency.

Cascading principles	Barriers	How cascading is prevented	Hindered loops	Occurrences in literature	Supporting literature
Appropriate Fit	1-Systemic constraints to effective circular pathways	Resale constraints – limited human resources, inadequate digital and logistical systems, and insufficient capacity for collection and sorting	Inner loops	1	Zanjirani Farahani et al. (2022)
		Remaking constraints – lack of technical skills and high processing costs	Outer loops	1	Paras et al. (2019)
		Integrated challenges in recycling systems, including technology, labor, skills, data, and financial motivation	Outer loops	5	Boschmeier et al. (2024); Charnley et al. (2024); Hornbuckle et al. (2023); Reike et al. (2023); Schmutz and Som (2022)
	2-Market fit and demand-side complexities	Non-conformity of recycled materials with quality/brand standards	Outer loops	2	Daddi et al. (2019); Reike et al. (2023)
Augmentation	1-Misaligned policies and trade restrictions	Non-adaptive tax policies – higher taxes on clothing repairs	Inner loops	1	Brändström et al. (2024)
		Tariffs and bans on the trade of second-hand clothing	Inner loops	1	Charnley et al. (2024)
Consecutive Relinking	1-System-level constraints and operational complexities in recycling	Recycling constraints – related to waste collection; infrastructure; technology; sufficient volumes for cost-effective processing; operational complexities	Outer loops	4	Maher et al. (2023); Morana and Seuring (2011); Salmi and Kaipia (2022); Schmutz and Som (2022)
	2-Regulatory shortcomings in recycling systems	Lack of harmonized regulations on recycled content as well as hygiene standards in material reuse	Outer loops	2	Brändström et al. (2024); Daddi et al. (2019)
Balancing Resource Metabolism	1-Infrastructure gaps and cost inefficiencies in recycling operations	Recycling constraints – infrastructure deficiencies; material sourcing challenges; price-driven limitations	Outer loops	3	Brändström et al. (2024); Maher et al. (2023); Schmutz and Som (2022)

Data availability

Data will be made available on request.

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